

MATH 4162/Math 6201

Numerical methods for differential equations

Fall 2025

Homework problem set #1

Show all necessary steps of working in the given space

Name: _____

ID: _____

Upload the assignment solution no later than

23:59 on Sep 26, 2025

Full marks 50

1. (based on week 1)[15 pts]

(a) Consider the KdV equation on $\mathbb{R}$

$$\frac{\partial u}{\partial t} - 6u \frac{\partial u}{\partial x} + \frac{\partial^3 u}{\partial x^3} = 0$$

with decay conditions $u, u_x, u_{xx} \rightarrow 0$ as $|x| \rightarrow \infty$ for each fixed t . Show that the “mass”

$$M(t) = \int_{-\infty}^{\infty} u(x, t) dx$$

is constant in time.

(b) Use the Fourier transform pair given as

$$\hat{u}(k, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} u(x, t) e^{ikx} dx \quad \text{and} \quad u(x, t) = \int_{-\infty}^{\infty} \hat{u}(k, t) e^{-ikx} dk.$$

and consider

$$\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^3 u}{\partial x^3}, \quad x \in \mathbb{R}, \quad t > 0,$$

with $u(x, 0) = f(x)$ and $u, \partial_x^m u \rightarrow 0$ as $|x| \rightarrow \infty$.

Convert the PDE into a system of ODE and finally, using inverse Fourier transform, show that

$$u(x, t) = \int_{-\infty}^{\infty} e^{(ik - k^2 + ik^3)t} \hat{f}(k) e^{-ikx} dk.$$

Q1(a)

$$M(t) = \int_{-\infty}^{\infty} u \, dx$$

$$\frac{\partial M}{\partial t} = \int_{-\infty}^{\infty} \frac{\partial u}{\partial t} \, dx = \int_{-\infty}^{\infty} (6uu_x - u_{xxx}) \, dx$$

$$\int_{-\infty}^{\infty} 6uu_x \, dx = 3 \int_{-\infty}^{\infty} \frac{\partial u^2}{\partial x} \, dx$$

$$= 3u^2 \Big|_{-\infty}^{\infty} = 0$$

$$\int_{-\infty}^{\infty} u_{xxx} \, dx = u_{xx} \Big|_{-\infty}^{\infty} = 0$$

$$\text{Hence } \frac{\partial M}{\partial t} = 0$$

$M(t)$ is constant in time.

Q1(b)

$$u_t + u_x = u_{xx} + u_{xxx}$$

$$\text{let } \hat{u}(k, t) = \mathcal{F}\{u\}$$

$$\text{then } \mathcal{F}\left\{\frac{\partial u}{\partial x}\right\} = -ik\hat{u}$$

$$\mathcal{F}\left\{\frac{\partial^2 u}{\partial x^2}\right\} = (-i)^2 k^2 \hat{u} = -k^2 \hat{u}$$

$$\mathcal{F}\left\{\frac{\partial^3 u}{\partial x^3}\right\} = (-i)^3 k^3 \hat{u} = i k^3 \hat{u}$$

$$\frac{\partial \hat{u}}{\partial t} - ik\hat{u} = -k^2 \hat{u} + ik^3 \hat{u}$$

$$\hat{u}(k, t) = e^{(ik - k^2 + ik^3)t} \hat{u}(k, 0)$$

$\nearrow \hat{f}(k)$

2. (based on week 2) [20 pts] The 1D wave equation is

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, \quad x \in \mathbb{R}, \quad t \geq 0.$$

- Derive the dispersion relation $\omega = \omega(k)$ for the wave equation, compute the phase speed c_p and the group speed c_g , state whether the PDE is dispersive and/or dissipative, and justify your answer briefly.
- Using Taylor series expansion about x_j , derive a five-point central finite difference approximation of $\frac{\partial^2 u}{\partial x^2}$ using the five-point stencil $\{x_{j-2}, x_{j-1}, x_j, x_{j+1}, x_{j+2}\}$ and report the lead error term.
- Derive the dispersion relation of the wave equation discretized in space with three-point stencil and provide a plot by comparing the exact dispersion relation with the numerical dispersion relation.

(a) $u = e^{i(kx + \omega t)}$

$$\frac{\partial^2 u}{\partial t^2} = i^2 \omega^2 e^{i(kx + \omega t)}$$

$$\frac{\partial^2 u}{\partial x^2} = i^2 k^2 e^{i(kx + \omega t)}$$

$$\Rightarrow i^2 \omega^2 = i^2 k^2$$

$$\boxed{\omega(k) = \pm k}$$

$$c_p = \frac{\omega(k)}{k} = \pm 1$$

$$c_g = \frac{\partial \omega(k)}{\partial k} = \pm 1$$

Since c_p does not depend on k ,

PDE is non-dispersive

$$\text{Now } \left| e^{i(kx + \omega t)} \right| \leq \left| e^{i(\operatorname{Re} \omega + i \operatorname{Im} \omega)t} \right|$$

$$\leq e^{-\operatorname{Im} \omega t}$$

$$\operatorname{Im} \omega = 0 \Rightarrow \text{no-dissipative}$$

2(b) Five point stencil

$$u_{j \pm 1} = u_j \pm \Delta x \frac{\partial u}{\partial x} \Big|_j + \frac{\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} \Big|_j \pm \frac{\Delta x^3}{3!} \frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^4}{4!} \frac{\partial^4 u}{\partial x^4}$$

$$u_{j \pm 2} = u_j \pm 2\Delta x \frac{\partial u}{\partial x} \Big|_j + \frac{4\Delta x^2}{2} \frac{\partial^2 u}{\partial x^2} \Big|_j \pm \frac{16\Delta x^3}{3!} \frac{\partial^3 u}{\partial x^3} + \frac{64\Delta x^4}{4!} \frac{\partial^4 u}{\partial x^4}$$

we want

$$\frac{\partial^2 u}{\partial x^2} = a u_{j+2} + b u_{j+1} + c u_j + d u_{j-1} + e u_{j-2}$$

Stencil is symmetric

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} &= \underline{a} u_{j+2} + \underline{b} u_{j+1} + c u_j + \underline{b} u_{j-1} + \underline{a} u_{j-2} \\ &= a(u_{j+2} + u_{j-2}) + b(u_{j+1} + u_{j-1}) + c u_j \end{aligned}$$

$$\frac{\partial^2 u}{\partial x^2} = (2a + 2b + c) u_j + \left(2a \frac{2\Delta x^2}{2} + 2b \frac{\Delta x^2}{2} \right) \frac{\partial^2 u}{\partial x^2}$$

$$+ \left(2a \frac{16}{4!} \Delta x^4 + 2b \frac{1}{4!} \Delta x^4 \right) \frac{\partial^4 u}{\partial x^4} + \dots$$

want

$$2a + 2b + c = 0 \quad (1)$$

$$(4a + b) \Delta x^2 = 1 \quad (2)$$

$$\left(\frac{4a}{3} + \frac{b}{12} \right) \Delta x^4 = 0 \quad (3)$$

$$(3) \Rightarrow b = -\frac{4a}{3}(12) = -16a$$

$$\text{Plug into (2)} \Rightarrow (4a - 16a) \Delta x^2 = 1, \quad a = \frac{1}{-12\Delta x^2}$$

$$\Rightarrow b = -16a = \frac{16}{12\Delta x^2} = \frac{4}{3\Delta x^2}$$

$$c = -2a - 2b = \frac{1}{6\Delta x^2} - \frac{8}{3\Delta x^2} = -\frac{5}{2\Delta x^2}$$

Hence $\frac{\tau_n}{\Delta x^v} = \frac{-u_{j+2} + 16u_{j+1} - 30u_j + 16u_{j-1} - u_{j-2}}{12\Delta x^v}$

leading order error

$$= C_1 \Delta x^4 \frac{\partial^6 u}{\partial x^6}$$

$$= O(\Delta x^4)$$

It is not necessary to know the value of C_1 , which is just a constant.

2(c)

3-point

$$u_{xx} = \frac{u_{j+1} - 2u_j + u_{j-1}}{\Delta x^2}, \quad \text{note, } u_{j+1} = u_{j+1}(t)$$

$$\text{Set } u(x_j, t) = e^{i(kj\Delta x + \omega t)}$$

$$\frac{\partial^2 u}{\partial t^2} = i^2 \omega^2 e^{i(kj\Delta x + \omega t)}$$

First check

$$\frac{u_{j+1} - 2u_j + u_{j-1}}{\Delta x^2}$$

$$= \frac{e^{i(kj\Delta x + \omega t)} e^{ik\Delta x} - 2e^{i(kj\Delta x + \omega t)} + e^{i(kj\Delta x + \omega t)} e^{-ik\Delta x}}{\Delta x^2}$$

Take $e^{i(kj\Delta x + \omega t)}$ out

$$i^2 \omega^2 = \frac{e^{ik\Delta x} - 2 + e^{-ik\Delta x}}{\Delta x^2}$$

$$-\omega^2 = \frac{2\cos k\Delta x - 2}{\Delta x^2} = -\frac{4\sin^2 \frac{k\Delta x}{2}}{\Delta x^2}$$

$$\omega = \frac{2\sin \frac{k\Delta x}{2}}{\Delta x}$$

In addition, we see that

$$c_p \frac{\omega}{k} = \frac{2\sin \frac{k\Delta x}{2}}{k\Delta x}$$

depends on $k \Rightarrow$ dispersion

$$\text{Exact } c_p = \frac{\omega}{k} = \pm 1 \Rightarrow \text{non dispersion}$$

3. [10pts] Consider the function

$$u(x) = e^{-\frac{x^2}{2\sigma^2}}, \quad x \in [-1, 1], \quad \sigma = 0.02.$$

- (a) Present a plot comparing exact $\frac{\partial u}{\partial x}$ with its numerical approximation.
- (b) How many grid points are needed for 2-norm error to be less than 10^{-2} ?

See the code

by error and trial approach

more than 13400 points

are needed

